

TITLE PAGE

Problem Statement ID – SIH26157

**Problem Statement Title - Supervisory
Analytics Tool for SOC Assessment
(SAT-SA)**

Theme - Blockchain & Cybersecurity

PS Category - Software

Team ID - 137529

Team Name - Blockers_1





THE PROBLEM



Currently, oversight bodies like NCIIPC struggle to independently verify SOC capabilities at scale. The process relies heavily on manual review and self-reported KPI dashboards that fail to highlight actual operational gaps.



CORE ISSUES



Dependence on Self-Reporting: Prevents independent verification; hides structural flaws.
Manual Review Bottlenecks: Periodic data oversight is unscalable; true anomalies are missed.



SILENT & LAZY GAPS



Silent Critical Assets: Blind spots where important infrastructure generates zero telemetry.
Analyst Inefficiency: Missed "lazy" behavior like <90s closures or copied investigation text.

TECHNICAL APPROACH & WORKFLOW (OFFLINE BATCH-FIRST) :

1



STEP 1: Input

Securely ingests periodic, offline SOC data (CSV, JSON, DB formats).



2



STEP 2: Evidence Graph Mapping

Maps "Expected Evidence Graph" (Alert → Case → Escalation) for independent verification.



3

STEP 3: Execution Gaps Detection

Automatically flags lazy analysis and investigation anomalies.



4



STEP 4: Negative Space Analysis

Identifies CII components with zero telemetry or logging activity.



5



STEP 5: Action

Calculates Supervisory Priority Score to queue only the most suspicious cases.



Our Solution

An offline, batch-first analytics platform enabling automated filtering without relying on self-reported KPI metrics.



Ultimate Outcome

Radically scales NCIIPC's oversight by empowering human examiners to focus purely on true, high-priority security anomalies.

TECHNICAL APPROACH

Tech Stack

FRONTEND



Backend & API



Data Analytics & Processing



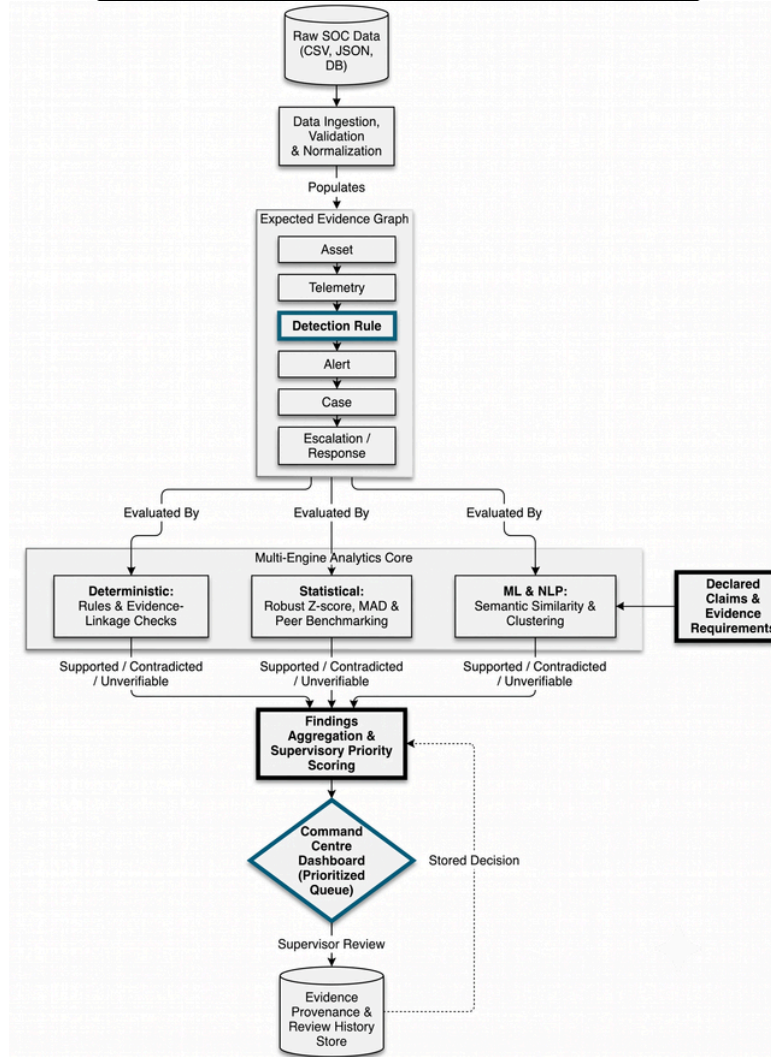
AI / ML / NLP



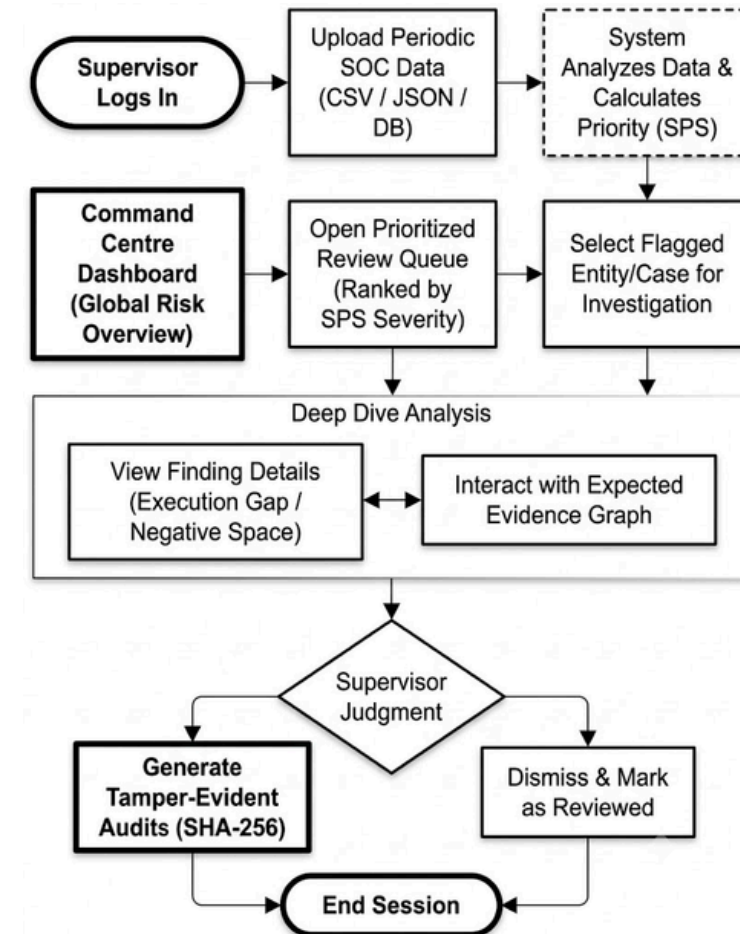
Database



System Architecture

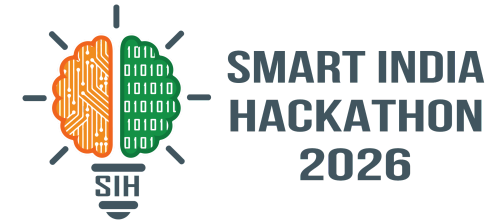


Flow Diagram



Your Team
Name

FEASIBILITY AND VIABILITY



Strict Scope Compliance

Built strictly as a supervisory analytics tool, not an operational SIEM. Ingests periodic metadata and workflows (CSV/JSON/DB), avoiding raw log/PCAP storage bottlenecks.

Hardware & Compute Feasibility

Lightweight open-source models (FAISS, scikit-learn) run on standard NCIIPC on-premise hardware.

On-premise
hardware

No GPU clusters

Lightweight
models

Explainability & Auditability

Every finding details:

- 1 WHAT happened
- 2 WHY it is suspicious
- 3 WHAT evidence supports it
- 4 WHAT benchmark was used

HYBRID AI & ANALYTICS PIPELINE (LOCAL PROCESSING)

**Deterministic
Python / Polars**
Rules engine · rapid closures · broken chains

**Statistical
Robust Z-score, MAD, IQR**
Peer benchmarking · metric-evidence divergence

**Machine Learning
Isolation Forest, DBSCAN**
Entity & case clustering · anomaly detection

**NLP
Local sentence-transformers**
Text similarity · templated, low-effort flags

Tamper-Evident Audit Ledger

Cryptographic hash chain, no heavy blockchain network:

Dataset Hash

Rule Version

Finding Hash

Continuous Validation Loop

Outputs cross-validated against historical expert manual review samples.

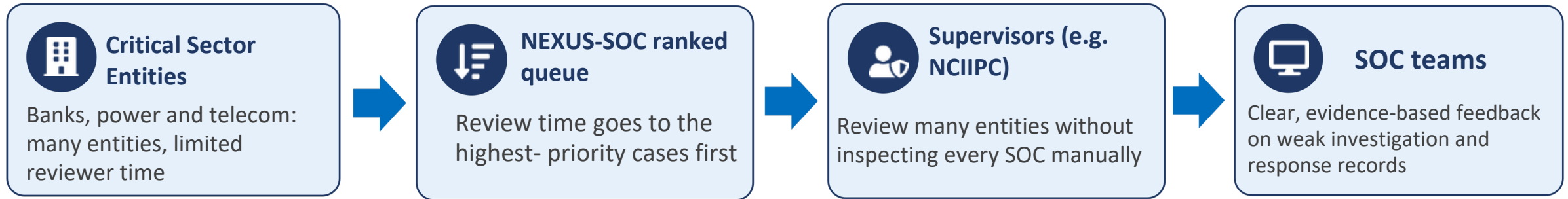
System Output

Expert Review Samples

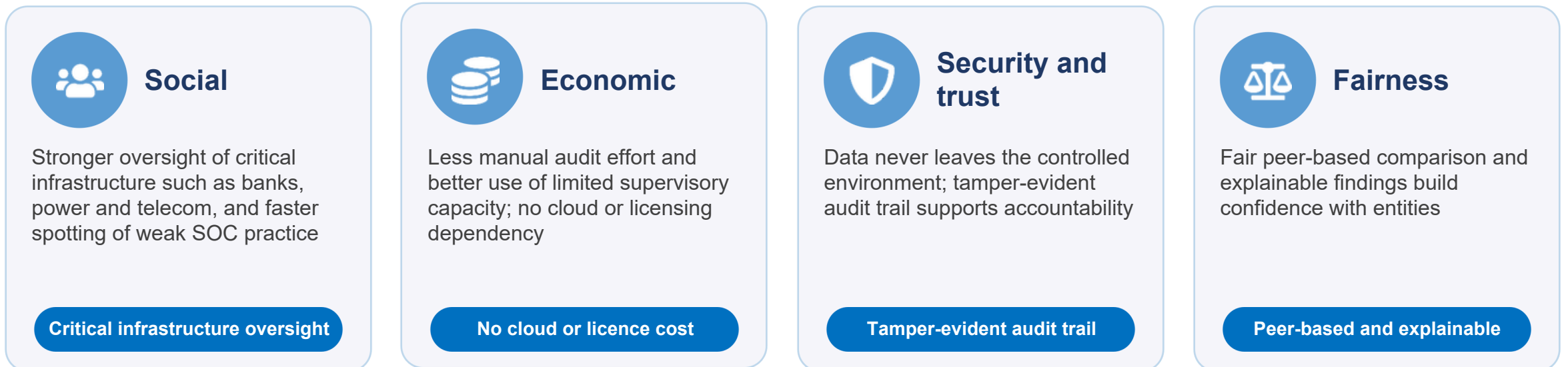
Validation & Model
Tuning

IMPACT AND BENEFITS

POTENTIAL IMPACT ON THE TARGET AUDIENCE :



BENEFITS OF THE SOLUTION :



Standards, Frameworks & Government Guidelines

- NCIIPC Guidelines for Protection of Critical Information Infrastructure: The primary regulatory baseline establishing the mandatory cyber resilience and auditing requirements for Critical Sector Entities (CSEs) in India.
- NIST SP 800-61 Rev. 2 (Computer Security Incident Handling Guide): Serves as the foundational blueprint for mapping the Expected Evidence Graph (Alert → Case → Escalation → Response).
- MITRE ATT&CK Framework: Utilized to map alert categories and verify detection coverage, enabling the Negative Space Engine to identify missing telemetry for critical systems.
- SOC-CMM (SOC Capability Maturity Model): Provides the peer-benchmarking criteria to assess whether a SOC's documented claims align with its actual operational maturity and evidence.

Production Link : <https://nexus-peach-three-19.vercel.app>

Technical USP References: Models & Algorithms

- Isolation-Based Anomaly Detection (Liu, Ting & Zhou, 2008 - IEEE ICDM): Research on the Isolation Forest algorithm, utilized for unsupervised anomaly detection of unusual operational patterns and rapid case closures without relying on pre-labeled data.
- Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks (Reimers & Gurevych, 2019 - EMNLP): The core NLP architecture deployed locally in the air-gapped environment to generate dense vector embeddings for SOC investigation text.
- A Density-Based Algorithm for Discovering Clusters (DBSCAN) (Ester et al., 1996 - KDD): The clustering algorithm used in tandem with Sentence-BERT to detect densely packed, highly similar investigation texts, flagging "Lazy Analysts" using copy-pasted templates.
- Billion-scale similarity search with GPUs (FAISS) (Johnson et al., 2017 - Meta AI): The vector similarity search library utilized to efficiently index and query text embeddings entirely offline.

YT Demo + Github : <https://github.com/kamble10pratik10-ops/Nexus>

Analytics & Statistical Methodologies

- Detecting outliers: Do not use standard deviation around the mean, use absolute deviation around the median (Leys et al., 2013): Academic justification for utilizing Median Absolute Deviation (MAD) and robust Z-scores instead of naive averages to prevent outlier CSEs from skewing peer-benchmarked KPIs.
- Polars: Blazingly Fast DataFrames: The high-performance, multithreaded Rust-based engine used for executing deterministic Evidence-Linkage checks on massive CSV and JSON batch datasets.